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Big Data-Driven Green Transformation: Bibliometrics and Text Mining on Water Pollution Reduction Practices

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Abstract: Water pollution poses a severe threat to ecological integrity, human health, and sustainable development globally. With the exponential growth of data from diverse sources such as remote sensing, environmental monitoring sensors, and computational models, big data has emerged as a transformative tool to address water pollution challenges. By using a new Bibliometrics and Text Mining tool (LitTopicMiner), this review synthesizes recent academic literature (1995–2025) to explore how big data analytics, including machine learning, remote sensing, and integrated modeling, enhances water pollution monitoring, source identification, risk assessment, and mitigation strategies. The findings highlight the multifaceted applications of big data in improving the accuracy of pollutant detection, optimizing pollution control measures, and supporting evidence-based decision-making. Additionally, the review discusses current limitations and future directions for leveraging big data to achieve more effective water pollution reduction.

Keywords: Water pollution; Big data; Bibliometrics; Text Mining

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1 Introduction

Water pollution, driven by anthropogenic activities such as industrial discharge, agricultural runoff, and urbanization, continues to degrade water quality worldwide (Fang et al., 2016; Prishchepenko et al., 2023). Traditional water pollution management approaches often suffer from limitations in real-time monitoring, spatial-temporal coverage, and complex pollutant source tracing. In recent years, big data technologies have revolutionized environmental management by enabling the integration and analysis of large-scale, heterogeneous datasets (McCabe et al., 2017). Big data, characterized by its volume, velocity, variety, and veracity, provides unprecedented opportunities to understand pollution dynamics, predict contamination events, and implement targeted mitigation actions (Xie et al., 2020). This review aims to systematically examine the role of big data in reducing water pollution, focusing on key applications, methodologies, and empirical findings from peer-reviewed literature.

2 Theoretical and literature foundation

2.1 Remote Sensing and Earth Observation

Remote sensing technologies, including satellite imagery and unmanned aerial vehicles (UAVs), have become primary data sources for large-scale water pollution monitoring. McCabe et al. (2017) emphasized that satellite missions (e.g., Landsat, GRACE) and UAVs provide high-resolution data on water quality parameters such as turbidity, nutrient concentrations, and algal blooms. For instance, Bojanowski et al. (2024) used satellite-derived climate data and the SWAT model to track nutrient loads in the Baltic Sea region, demonstrating that remote sensing data enhances the spatial-temporal accuracy of pollution monitoring. Similarly, Naboni and Garcia (2017) developed architectural devices integrated with remote sensing tools to monitor thermal pollution and water flows in tropical regions, highlighting the potential of combined data sources for localized pollution assessment.

2.2 Sensor Networks and IoT

Internet of Things (IoT) sensor networks enable real-time, in-situ monitoring of water quality. Arshad et al. (2024) designed an IoT-based system with body area sensors to monitor livestock-related pollution, where physiological parameters of cattle are linked to nutrient runoff risks. Elewa et al. (2025) utilized sensor data to optimize the performance of geopolymer adsorbents for heavy metal removal, showing that real-time sensor data improves the efficiency of pollution control technologies. Additionally, Radulescu (2020) integrated sensor data with computational fluid dynamics models to analyze thermal pollution from industrial processes, emphasizing the role of high-frequency sensor data in understanding pollutant transport.

2.3 Computational Models and Simulation Data

Computational models generate large volumes of data for pollution prediction and risk assessment. Amador et al. (2025) developed a spatially explicit modeling chain (erahumed-RiceWQ-Delft3D) to simulate pesticide transport in Mediterranean wetlands, using big data analytics to integrate hydrological, agricultural, and hydrodynamic datasets. Wang et al. (2024) applied machine learning-based quantitative structure-activity relationship (QSAR) models to screen persistent, bioaccumulative, and toxic (PBT) chemicals, demonstrating that computational models process large chemical datasets to identify high-risk pollutants. Furthermore, Lazartigues et al. (2013) used principal component analysis and hierarchical cluster analysis to analyze pesticide contamination data in fish ponds, illustrating how statistical modeling of big data reveals pollution patterns.

2.4 Pollutant Source Identification

Big data analytics facilitates precise identification of pollution sources, a critical step for targeted mitigation. Subbotin et al. (2017) used logical-linguistic modeling to analyze nitrogen loss data from manure utilization, identifying key factors (e.g., manure processing type, soil application method) contributing to agricultural pollution. In the Baltic Sea region, Bojanowski

et al. (2024) identified agricultural lands and forest areas as major nutrient sources under climate change scenarios, using big data to disaggregate pollution contributions across subcatchments. For urban pollution, Bolca et al. (2007) employed remote sensing and GIS data to link urbanization to wetland degradation and water pollution in Turkey's Balçova Delta, showing that spatial big data reveals land-use-driven pollution pathways.

2.5 Risk Assessment and Prediction

Big data enables accurate risk assessment and prediction of water pollution events. Charrasse et al. (2023) developed the MITOMATICS pipeline, combining high-content screening and big data analytics to assess pesticide-induced mitochondrial damage, providing early warnings of ecological risks. In the Gulf of Finland, Prishchepenko et al. (2023) analyzed historical geochemical data to predict heavy metal accumulation trends, demonstrating that long-term big data records improve pollution risk forecasting. Additionally, Ahamad et al. (2024) used questionnaire survey data and statistical analysis to assess the impact of manufacturing industries on water quality in Bangladesh, showing that big data from stakeholder perspectives complements environmental monitoring data for comprehensive risk assessment.

2.6 Mitigation Strategy Optimization

Big data optimizes pollution mitigation strategies by identifying cost-effective and efficient interventions. Li et al. (2021) applied life cycle assessment and big data simulation to optimize fertilization schemes in South China, reducing eutrophication and acidification while maintaining crop yields. Elewa et al. (2025) used big data from adsorption experiments to optimize geopolymer adsorbents for heavy metal removal, achieving nearly 100% removal efficiency for Cr, Cd, and Pb. For agricultural pollution, Lazartigues et al. (2013) recommended reducing pesticide application and improving crop rotation based on big data analysis of fish contamination, highlighting the role of data-driven management in sustainable agriculture.

3 Research method

3.1 Research Design

This study is a literature review by using bibliometric and text mining analyses to examine the impact of big data on water pollution reductions. A new bibliometric and text mining tool (LitTopicMiner) has been developed in this study. This new tool has combined traditional bibliometric analyses with Latent Dirichlet Allocation (LDA). LDA is an unsupervised probabilistic topic modeling algorithm that has been applied to extract latent research topics from English abstracts of the collected literature. LitTopicMiner has been implemented in Python.

3.2 Data Collection

Bibliometric data were sourced from the Web of Science database. The inclusion criteria encompassed peer-reviewed articles published in English from 1992 to 2026, focusing on key themes on big data and water pollution. A total of 547 articles were identified, providing a robust dataset for subsequent analysis.

3.3 LitTopicMiner

LitTopicMiner is a custom-developed Python program designed for systematic bibliometric analysis and unsupervised Latent Dirichlet Allocation (LDA) topic modeling of scientific literature, tailored specifically for Web of Science (WOS) Bibtex full-record datasets. The core design rationale of the program adheres to three key principles: data integrity and accuracy, functional integration and scalability, and scholarly usability and standardization, aiming to address the pain points of fragmented bibliometric analysis and non-specialized topic modeling in existing tools, and to provide a one-stop, reproducible analytical workflow for literature research in various disciplines.

The program's architecture is modularized into two core classes and a set of auxiliary functional modules, with a clear separation of data parsing, preprocessing, analytical computation, and result visualization to ensure code readability and maintainability. The BibtexParser class is engineered to solve the common issues of WOS Bibtex file parsing, including automatic encoding detection (supporting utf-8, gbk, gb2312, latin-1) to avoid decoding errors, and a refined regular

expression-based field extraction method to fix author name truncation and accurately extract multi-line fields such as affiliations and abstracts. It unifies field naming and merges author-defined and WOS-supplemented keywords, while filling missing core fields with empty strings to ensure data type consistency, laying a solid foundation for subsequent analysis.

The LiteratureAnalyzer class serves as the program's core execution module, integrating comprehensive bibliometric analysis and LDA topic modeling with a highly optimized data preprocessing pipeline. For bibliometric analysis, it implements rule-based precise extraction of corresponding author affiliations (extracting the complete university name between the first and second commas after the (Corresponding Author) tag) and full author name parsing (converting WOS's last name, initials format to complete names), effectively filtering redundant institutional information and ensuring the accuracy of author and institutional metrics. It supports multi-dimensional bibliometric analysis including publication trend statistics, high-frequency keyword/author/journal/institution identification, and co-occurrence network construction (author collaboration and keyword co-occurrence), with all analytical results saved as Excel files for traceability.

For LDA topic modeling, the program embeds a rigorous text preprocessing workflow for English abstracts—including text normalization, stop word and ultra-short word filtering, and WordNet lemmatization—to eliminate semantic noise and unify lexical forms. It adopts the Bag-of-Words model with optimized CountVectorizer parameters (max_df=0.8, min_df=2) to construct a high-quality document-term frequency matrix, filtering high-frequency universal words and low-frequency rare words to improve feature discriminability. The LDA model is implemented with batch learning for stable convergence, supporting both manual topic number setting and automatic optimal topic number selection via 3-fold cross-validation grid search (based on perplexity minimization). It extracts the top 10 core words for each latent topic and assigns the optimal topic to each document by maximum probability, with quantitative results (topic-word weights, document-topic distribution) stored in a dedicated Excel file.

The program features a user-centric design with minimal configuration requirements—users only need to modify three parameters (Bibtex file path, output directory, journal format) to execute the full workflow. It provides a built-in journal format adaptation module (supporting Nature, MDPI, PLOS ONE, Science), automatically adjusting figure DPI, size, and layout to meet academic journal submission requirements. All visualizations (trend charts, bar charts, network graphs, LDA topic plots) are generated with matplotlib and seaborn, with Chinese and English dual-language support for titles and labels, and all results (charts and Excel data) are saved to a self-created output directory for easy management. Additionally, the program disables irrelevant warnings, adds detailed progress prompts, and includes error handling for empty datasets, ensuring a smooth user experience even for non-professional Python users.

4 Data analysis and results

4.1 Word cloud

By using LitTopicMiner, the word cloud of research key words is presented in Figure 1. This word cloud visually encapsulates the core thematic landscape of research on big data applications in water pollution reduction. Dominant terms—"water", "big data", "water pollution", and "quality"—occupy prominent positions (larger font sizes), reflecting their centrality as the field's foundational focus: integrating big data methodologies to address water pollution and quality management challenges.

Complementary keywords reveal the interdisciplinary and technical dimensions of this research domain. Methodological terms (e.g., model, analysis, machine learning, artificial intelligence, regression, algorithm) highlight the reliance on data-driven analytical and computational approaches to model, predict, or assess water systems. Thematic sub-foci (e.g., contamination, eutrophication, heavy metal, nutrient, pesticide) specify the key pollutants and water degradation issues targeted by big data applications. Contextual and operational terms (e.g., monitoring, management, system, watershed, urban, agricultural) situate the research within real-world environmental governance frameworks, spanning both urban and agricultural water systems, as well as watershed-scale management. Supporting concepts (e.g., environment, sustainability, impact, risk assessment, removal) emphasize the field's dual goals: understanding pollution impacts and developing data-

informed strategies for mitigation and sustainable water resource management.

Notably, terms like IoT, remote sensing, neural network signal the integration of emerging technologies with big data, while geographic/ecosystem references (river, lake, groundwater, marine) reflect the breadth of water bodies studied. Collectively, the word cloud illustrates that this research area is characterized by a convergence of big data analytics, environmental science, and practical water management, with a focus on technical innovation to address diverse water pollution challenges.

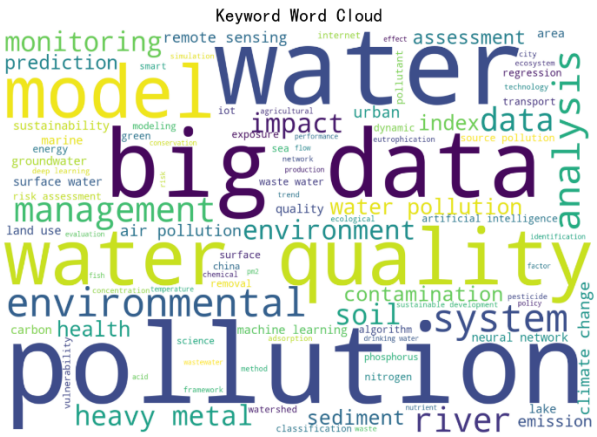

Figure 1. Word cloud

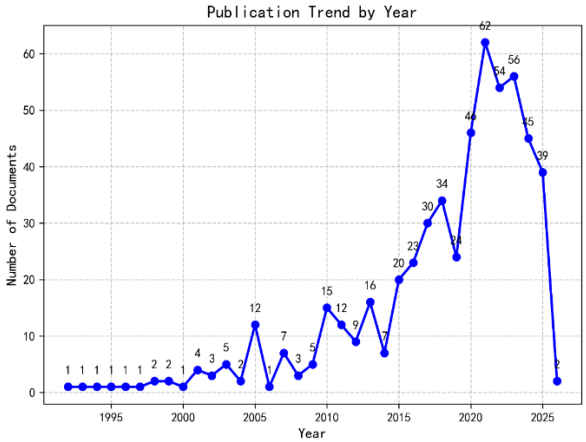

Figure 2. Publication trend

4.2 Publication trend

We synthesize the 35-year publication trend of research on big data applications in water pollution reduction, based on annual publication counts and growth rates from 1992 to 2025. The academic exploration of this interdisciplinary field exhibits a clear long-term upward trajectory, divided into three distinct phases, despite periodic fluctuations. Publication trend is presented in Figure 2.

The formative phase (1992–2004) was marked by minimal and unstable research output. Starting with 1 publication in 1992, the annual count remained static at 1 until 1997, reflecting limited initial attention to big data's environmental applications. Modest volatility emerged thereafter: a 100% increase to 2 publications in 1998, a 50% decline back to 1 in 2000, and a 300% surge to 4 in 2001, followed by fluctuations between 2 and 5 publications through 2004. This instability indicated the nascent stage of technological integration in water pollution research.

The expansion phase (2005–2015) signaled growing recognition of big data's potential. A pivotal 500% growth in 2005 (from 2 to 12 publications) marked the start of accelerated interest. While temporary setbacks occurred—such as a 91.67% decline to 1 publication in 2006—the overall trend was upward: a 600% increase to 7 in 2007, a 200% jump to 15 in 2010, and a 185.71% surge to 20 in 2015. This phase aligned with advancements in data analytics and rising global concern over water pollution, driving expanded interdisciplinary research.

The maturation phase (2016–2025) featured high, stable output. Annual publications grew steadily from 23 (2016) to a peak of 34 (2018), with a temporary dip to 24 (2019) followed by a 91.67% recovery to 46 (2020) and a 35-year high of 62 (2021). Fluctuations between 39 and 56 publications in 2022–2025 indicated stabilization, confirming the field's establishment as a core area of environmental research.

4.3 Research Organization Analysis

We examines the top 10 institutions contributing to research on big data applications in water pollution reduction, based on publication volume and academic influence indicators (Figure 3). The distribution of leading institutions reflects the global and interdisciplinary nature of this research domain, with representation across multiple continents and institutional types.

Big data analytics has emerged as a pivotal technological approach for water pollution reduction research, with a cluster of academic and research institutions globally leading the output of relevant scholarly literature. The Chinese Academy of

Sciences (Chinese Acad Sci) stands as the most prominent contributor, accounting for 18 publications and demonstrating its dominant position in integrating big data technologies with water pollution control research, which reflects the institution's robust research capacity in cross-disciplinary applications of data science and environmental engineering.

Beijing Normal University (Beijing Normal Univ) ranks second with 10 publications, emerging as a core academic force in this field among Chinese universities, with its research likely focusing on big data-driven water quality monitoring, pollution source identification and predictive modeling for water pollution reduction. The University of the Chinese Academy of Sciences (Univ Chinese Acad Sci) follows with 6 publications, forming a collaborative research ecosystem with the Chinese Academy of Sciences and further consolidating the research strength of the Chinese Academy of Sciences system in this interdisciplinary domain.

Indian Institute of Technology (Indian Inst Technol) is the leading non-Chinese institution with 5 publications, representing the important progress of big data applications in water pollution reduction research in India and becoming a key node of this research field in South Asia. A group of well-known universities and research institutions share the same publication volume of 4, including Zhejiang University (Zhejiang Univ), Russian Academy of Sciences (Russian Acad Sci), Nanjing University (Nanjing Univ) and Hohai University (Hohai Univ); these institutions cover both top comprehensive universities and specialized research bodies in hydrology and environmental science, enriching the research dimensions of big data in water pollution reduction from technical development, regional water pollution governance and hydrological data analysis.

With 3 publications each, University Politehnica Bucuresti (Univ Politehn Bucuresti) and Shanghai Jiao Tong University (Shanghai Jiao Tong Univ) serve as important supplementary forces, expanding the global geographical coverage and disciplinary diversity of the research field. Collectively, these institutions form the core research network of big data applications in water pollution reduction, with Chinese research institutions occupying the central position, while international institutions from South Asia, Europe and Russia provide cross-regional research perspectives, laying a solid foundation for the global development and practical application of big data-based water pollution reduction strategies.

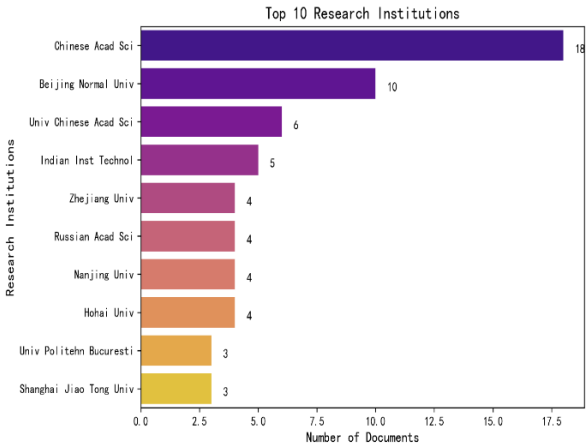

Figure 3. Research Organization

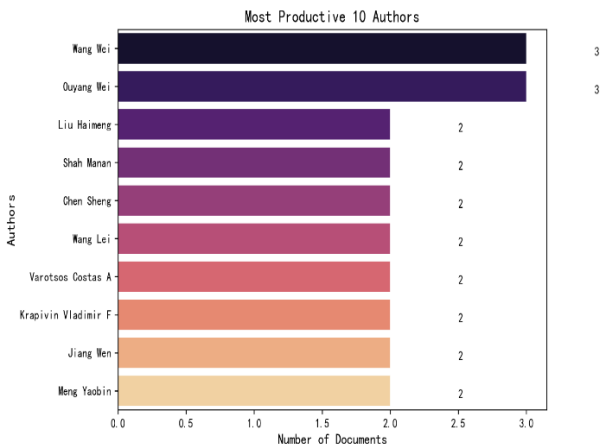

Figure 4. Author analysis

4.4 Author Analysis

We have identified the 10 most productive research in this field (Figure 4). Big data-driven research on water pollution reductions has formed a core group of contributing authors, with Wang Wei and Ouyang Wei standing out as the most productive ones, each publishing 3 relevant literatures. Their consistent outputs indicate their in-depth engagement and leading role in integrating big data technologies with water pollution control, covering potential research directions such as data-based pollution monitoring, simulation and reduction strategy optimization.

A second tier of key authors consists of ten scholars each with 2 publications, including Liu Haimeng, Shah Manan, Chen Sheng, Wang Lei, Varotsos Costas A, Krapivin Vladimir F, Jiang Wen and Meng Yaobin. These authors exhibit both regional diversity and disciplinary breadth—Chinese authors like Liu Haimeng, Chen Sheng and Jiang Wen enrich localized research

perspectives on water pollution reduction using big data, while international scholars such as Shah Manan, Varotsos Costas A and Krapivin Vladimir F provide cross-regional insights.

Collectively, these authors form the core research force in this field. The leading contributions of Wang Wei and Ouyang Wei, coupled with the supplementary efforts of the other eight scholars, promote the development and practical application of big data technologies in water pollution reduction, laying a foundation for further interdisciplinary and cross-regional research.

4.5 Journal analysis

Journals play a crucial role in disseminating research findings on big data applications in water pollution reductions, with a group of core journals dominating the publication output (Figure 5). Science of The Total Environment ranks first with 15 publications, emerging as the most influential platform in this field, which reflects its strong academic influence and focuses on interdisciplinary research integrating big data, environmental science and pollution control.

Sustainability follows closely with 14 publications, focusing on the sustainability-oriented application of big data in water pollution reduction, highlighting the combination of technological innovation and ecological sustainability. Three journals share the same publication volume of 12, namely Environmental Science and Pollution Research, Environmental Monitoring and Assessment, and Water. These journals cover key research directions such as pollution monitoring, assessment, and water resource management, providing important channels for related research dissemination.

Environmental Pollution and Journal of Cleaner Production contribute 9 and 8 publications respectively, focusing on pollution mechanism analysis and cleaner production-oriented water pollution reduction driven by big data. Additionally, Environment Development and Sustainability, Polish Journal of Environmental Studies, and Frontiers in Environmental Science each have 5 publications, serving as important supplements. Collectively, these journals form the core dissemination network, promoting the development and application of big data in water pollution reduction.

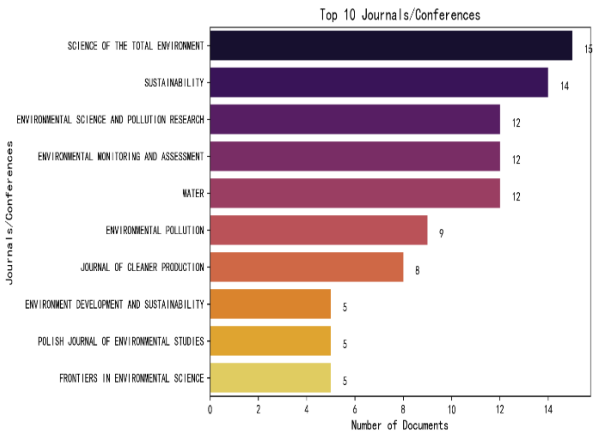

Figure 5. Journal analysis

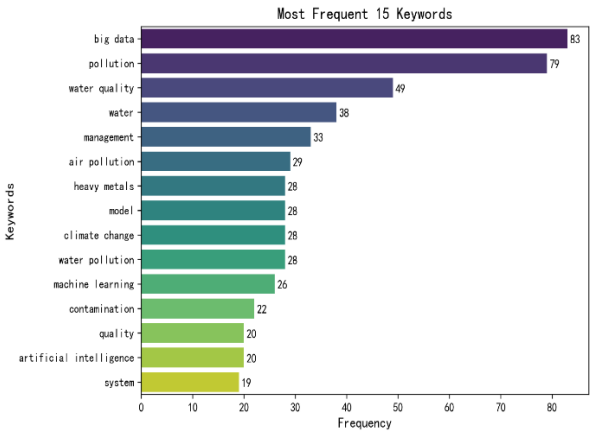

Figure 6. Keyword Analysis

4.6 Keyword Analysis

Keywords are the core embodiment of research hotspots and thematic focuses, which can intuitively reflect the research direction and content framework of big data applications in water pollution reductions (Figure 6). Based on the frequency statistics of relevant keywords, the research field presents obvious hierarchical characteristics and interdisciplinary attributes, with core keywords leading the research direction and secondary keywords enriching the research content, forming a complete thematic network.

Among all keywords, big data ranks first with a frequency of 83, which fully indicates that big data technology is the core driving force and core research object of this field, and all research contents are closely centered on the development, application and optimization of big data in water pollution reduction. Pollution follows closely with 79 occurrences, which is closely related to the core theme of water pollution reduction, reflecting that the research focuses on the identification, monitoring, control and governance of various pollution problems in the water environment. Water quality, with a frequency

of 49, is another core keyword, indicating that improving and ensuring water quality is the important goal of big data application in water pollution reduction, and relevant research pays great attention to the evaluation and improvement of water quality.

Water (38 times) and management (33 times) form the basic context of the research field, among which water defines the research scope, and management highlights the practical orientation of the research, that is, using big data technology to optimize water environment management and improve the efficiency of pollution reduction. A group of keywords with the same frequency of 28, including heavy metals, model, climate change and water pollution, reflect the multi-dimensional expansion of the research field: heavy metals point to the research on specific pollution sources; model indicates that big data is often combined with simulation models to carry out pollution prediction and governance scheme design; climate change reveals the cross correlation between water pollution reduction and global climate issues; water pollution directly points to the core research theme, further confirming the focus of the field.

Air pollution (29 times) shows that the research has a certain cross-domain attribute, focusing on the interaction between air and water pollution in the context of big data monitoring. Machine learning (26 times) and artificial intelligence (20 times) reflect the important technical support of intelligent algorithms for big data application in water pollution reduction, which is an important research hotspot in the field. In addition, contamination (22 times), quality (20 times) and system (19 times) supplement the research content from the aspects of pollution status description, quality evaluation and system construction. In general, these keywords form a thematic system with big data as the core, water pollution reduction as the goal, and technology, management and multi-environmental factors as the supplements, reflecting the interdisciplinary and practical characteristics of the research field.

4.7 Keyword network

This keyword co-occurrence network visualizes the interdisciplinary landscape of big data applications in water pollution reduction, revealing a densely interconnected knowledge ecosystem with thematic clusters that reflect both technical and environmental priorities (Figure 7). At the network's core, the node for "big data" occupies a central position, serving as a hub that bridges computational methodologies and environmental remediation goals. It exhibits strong co-occurrence with terms like "water quality", "machine learning", "remote sensing", and "data analytics", highlighting its foundational role in integrating real-time monitoring and predictive modeling to inform pollution mitigation strategies.

The network reveals three primary thematic clusters radiating from this central hub. First, a computational analytics cluster emerges, marked by nodes such as "artificial intelligence", "algorithms", and "optimization". These terms underscore the use of big data-driven predictive models to identify pollution sources, simulate contaminant transport, and optimize treatment processes. Co-occurrences with "Internet of Things" (IoT) and "sensor networks" further emphasize the role of real-time data acquisition in creating dynamic, adaptive monitoring systems that enable proactive rather than reactive pollution control.

Second, an environmental monitoring and remediation cluster is anchored by nodes including "water quality assessment", "heavy metals", "microplastics", and "persistent organic pollutants". Here, big data is linked to the quantification of emerging contaminants, spatial analysis of pollution hotspots, and the evaluation of remediation efficacy. Co-occurrences with "remote sensing" and "satellite imagery" highlight the integration of multi-scale data sources to monitor large aquatic ecosystems, from coastal estuaries to river basins, while terms like "adsorption" and "activated carbon" connect data analytics to the development of targeted treatment technologies.

Third, a sustainability and policy cluster is visible through nodes such as "sustainable development goals", "climate change", and "risk assessment". This cluster reflects how big data supports evidence-based policy-making by synthesizing long-term pollution trends with socio-economic factors, enabling stakeholders to design strategies that align with global water security targets. Co-occurrences with "resilience" and "urbanization" further underscore the role of data-driven approaches in addressing the compounding pressures of urban growth and climate variability on water systems.

Peripheral nodes, including "methylmercury", "biodiversity", and "ecotoxicity", indicate emerging research frontiers where big data is increasingly applied to understand the ecological impacts of pollution. Meanwhile, links to "public health" and "food security" demonstrate the network's extension into human dimensions, emphasizing the cascading benefits of data-

informed pollution reduction for both ecosystems and communities.

Overall, the network illustrates that big data functions not merely as a technical tool but as a transdisciplinary catalyst, uniting computational science, environmental engineering, and policy to advance integrated solutions for water pollution reduction. Its centrality and dense connectivity across clusters underscore the growing recognition of data-driven innovation as essential to achieving sustainable water management in the face of global environmental change.

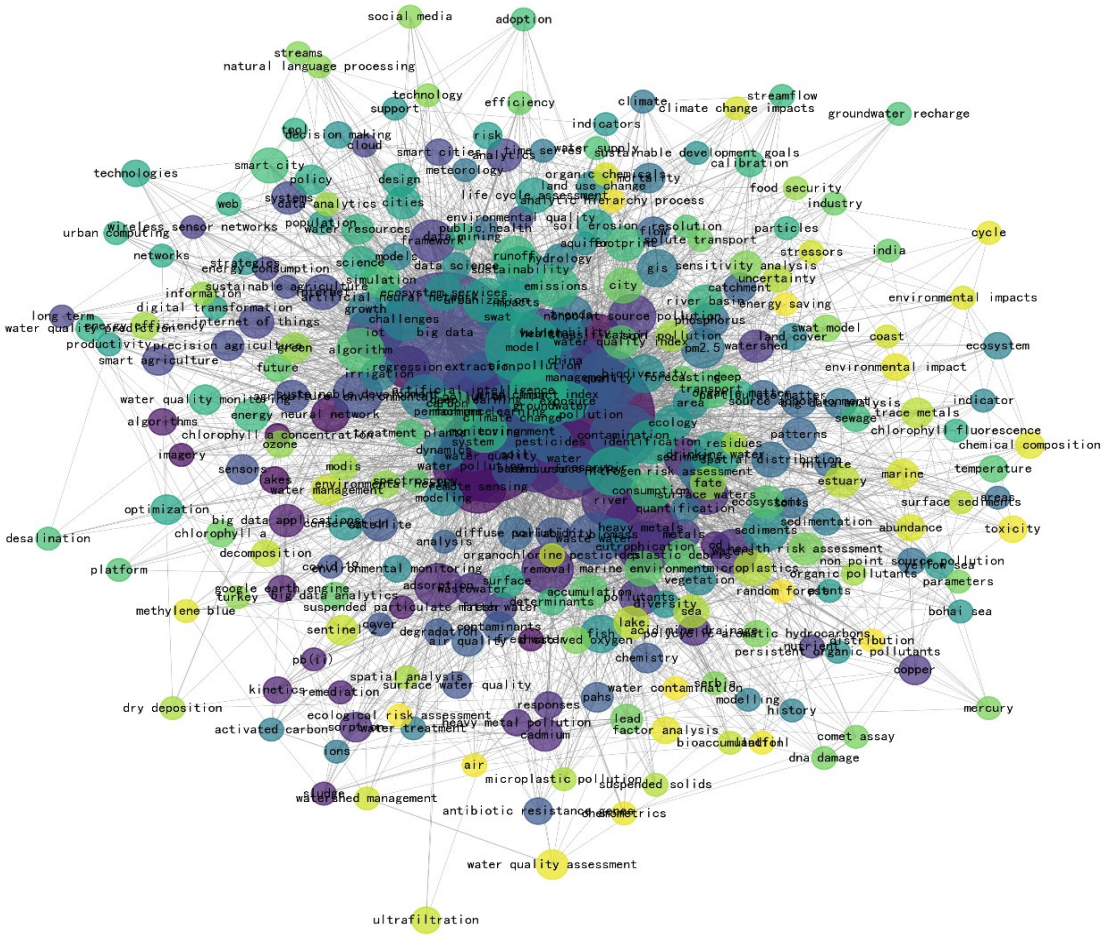

Figure 7. Keyword Co-occurrence network

4.8 Key temporal trend

The temporal keyword distribution analysis reveals a clear evolution in the academic focus on water pollution reduction, with big data emerging as a pivotal theme only in the most recent period (2010–2015). The temporal keyword distribution is presented in Figure 8.

In the earliest phase (1992–1997), research priorities centered on atmospheric and chemical processes related to pollution, with keywords such as “dry deposition”, “nitric acid”, and “ozone” dominating the landscape. This period showed no explicit mention of big data, reflecting a focus on field-based measurements and fundamental chemical interactions rather than data-driven analytics. The 1998–2003 phase shifted toward broader environmental systems, with terms like “watershed”, “ecosystems”, and “water quality” gaining prominence, but computational or data science approaches remained absent.

The 2004–2009 period marked a transition toward applied water pollution mitigation, as seen in the rise of keywords including “eutrophication”, “contamination”, and “removal”. While “water quality” and “pollution” became central topics, the discourse still lacked explicit references to big data, indicating reliance on traditional monitoring and modeling techniques rather than large-scale data integration.

The most dramatic shift occurred in the 2010–2015 period, where “big data” emerged as a distinct keyword with a notable frequency, alongside high-frequency terms like “pollution” and “water quality”. This coincided with the rise of complementary

themes such as “ultrafiltration”, “climate change”, and “impact”, signaling a convergence of data analytics, advanced treatment technologies, and climate resilience in water pollution research. The appearance of “big data” in this era reflects a paradigm shift: academic research began to leverage large, heterogeneous datasets to address complex pollution challenges, from identifying nonpoint sources to optimizing remediation strategies.

The temporal trends thus illustrate a progression from foundational chemical and ecological studies to a data-informed, solution-oriented approach, with big data positioned as a transformative tool in the latest phase. This evolution underscores the growing recognition of data-driven methodologies as essential for addressing the increasing complexity of water pollution in the context of global environmental change.

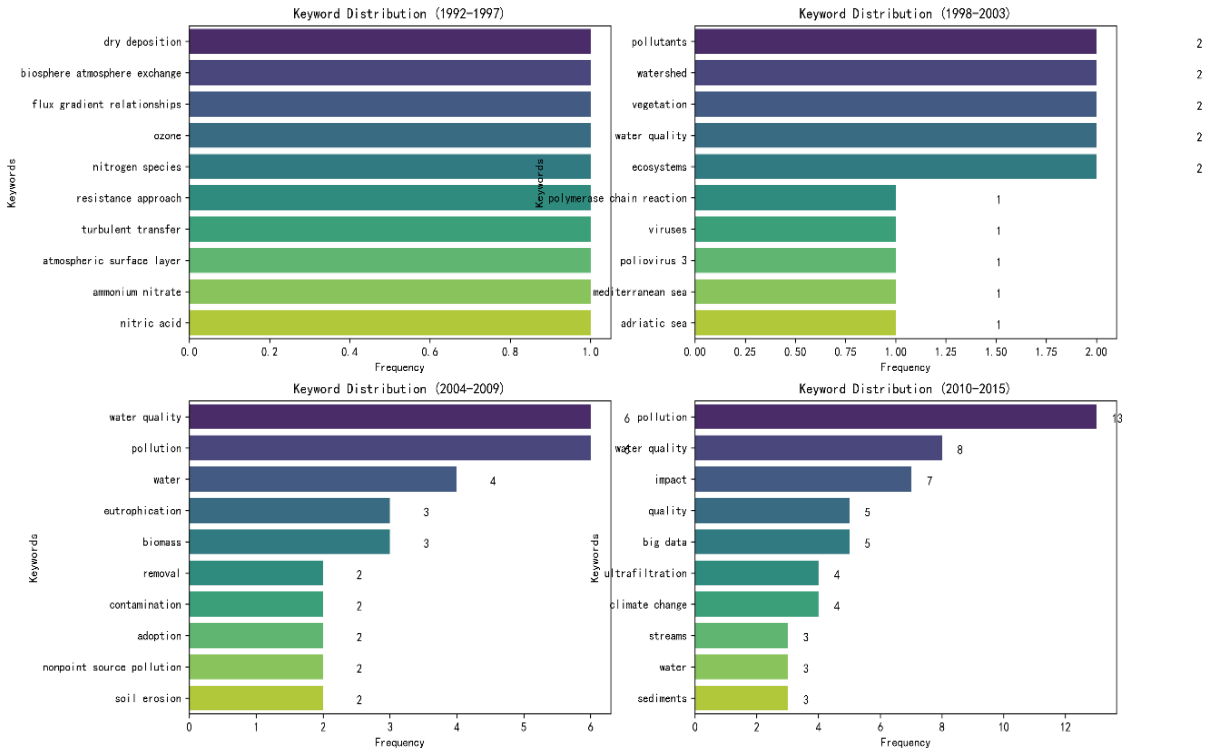

Figure 8. Key temporal trend

4.9 LDA analysis

This Latent Dirichlet Allocation (LDA) analysis identifies five distinct thematic clusters in the academic literature on big data applications for water pollution reduction, providing a structured view of the field’s intellectual landscape (Figure 9 and 10). The topic proportion distribution reveals that Topic 4 (23.2%) and Topic 5 (22.9%) are the most dominant, followed by Topic 2 (19.6%), Topic 3 (17.6%), and Topic 1 (16.8%). This hierarchy reflects the relative prevalence of different research priorities, from data-driven pollution monitoring to ecosystem-scale analysis, with big data serving as a cross-cutting analytical framework across all clusters.

Topic 1 (16.8%) centers on the spatial and chemical characterization of water pollution, with core keywords including “concentration”, “river”, “metal”, “heavy”, “site”, and “pollution”. The prominence of “concentration” (highest weight) underscores a focus on quantifying pollutant levels, particularly for heavy metals such as lead, cadmium, and mercury, in fluvial systems. Terms like “river”, “site”, and “sediment” highlight the emphasis on field-based sampling and spatial mapping of contamination hotspots, while “value”, “study”, and “result” reflect the empirical, data-intensive nature of this research. Big data implicitly supports this theme by enabling the integration of large datasets from multiple monitoring sites to identify spatial patterns and trace pollutant sources, such as industrial discharge or agricultural runoff. This topic aligns with foundational environmental chemistry research, where data analytics enhances the precision of contamination assessments and supports targeted remediation strategies.

Topic 2 (19.6%) represents the most explicit focus on big data methodologies, with core keywords including "quality", "monitoring", "pollution", "model", "time", "analysis", and "big". The dominance of "quality" (highest weight) and "monitoring" reflects a focus on real-time water quality tracking, while "big" and "analysis" highlight the use of large-scale data processing to derive actionable insights. Terms like "model", "time", and "method" underscore the development of predictive models and temporal trend analysis, which leverage big data to detect pollution events and forecast water quality changes. This theme encompasses the integration of sensor networks, remote sensing, and machine learning algorithms to create adaptive monitoring systems, addressing the limitations of traditional, periodic sampling. For example, real-time data from IoT sensors can be combined with historical datasets to identify anomalies, enabling rapid responses to pollution incidents. This topic thus positions big data as a transformative tool for enhancing the efficiency and responsiveness of water pollution management.

Topic 3 (17.6%) focuses on the intersection of big data, pollution reduction, and sustainable development, with core keywords including "environmental", "technology", "management", "environment", "study", "pollution", "research", "big", "development", and "sustainable". The highest-weighted term, "environmental", signals a holistic approach to pollution reduction that considers ecological integrity, while "sustainable" and "development" reflect alignment with global frameworks such as the UN Sustainable Development Goals (SDGs). Terms like "management" and "technology" highlight the role of big data in informing evidence-based policy and optimizing remediation technologies, such as advanced oxidation processes or bioremediation. This theme emphasizes the transdisciplinary nature of water pollution research, where data analytics bridges environmental science, engineering, and policy to design strategies that balance ecological health with socio-economic needs. For instance, big data can synthesize long-term pollution trends with demographic data to assess the equity of water access and target interventions in vulnerable communities.

Topic 4 (23.2%) is the most dominant cluster, focusing on urban water pollution and its environmental impacts, with core keywords including "pollution", "air", "environmental", "city", "big", "monitoring", "urban", "environment", "energy", and "quality". The highest-weighted term, "pollution", reflects a broad focus on urban contamination sources, including industrial emissions, stormwater runoff, and wastewater discharge, while "air" and "environmental" highlight the interconnectedness of water and air pollution in urban ecosystems. Terms like "city", "urban", and "monitoring" underscore the challenges of managing pollution in densely populated areas, where big data supports the integration of multi-modal datasets (e.g., traffic, energy use, and water quality) to identify cumulative impacts. This theme also addresses the role of big data in enhancing urban resilience, such as by modeling the effects of climate change on stormwater systems or optimizing green infrastructure to reduce pollution. The prominence of "big" and "monitoring" confirms the centrality of data-driven approaches in addressing the complexity of urban water pollution.

Topic 5 (22.9%) focuses on watershed-scale pollution dynamics and ecosystem interactions, with core keywords including "area", "model", "pollution", "soil", "study", "result", "quality", "land", "river", and "watershed". The highest-weighted term, "area", reflects a focus on spatial analysis of large hydrological units, while "watershed" and "river" emphasize the importance of landscape-scale processes in pollution transport. Terms like "soil", "land", and "model" highlight the use of big data to model the movement of pollutants from terrestrial to aquatic systems, such as sediment-bound nutrients or pesticides. This theme also includes the evaluation of best management practices (BMPs) for nonpoint source pollution, where data analytics compares the efficacy of strategies like riparian buffers or cover crops. Big data enables the integration of remote sensing imagery, soil surveys, and stream gauge data to create holistic watershed models, supporting adaptive management and ecosystem restoration efforts.

Across all topics, big data emerges as a unifying analytical framework that enhances the precision, scalability, and responsiveness of water pollution reduction efforts. The distribution of topics reflects a progression from foundational chemical characterization (Topic 1) to integrated, data-driven management (Topics 2–5), highlighting the field's evolution toward transdisciplinary, solution-oriented research. The dominance of Topics 4 and 5 underscores the growing focus on urban and watershed-scale challenges, which require large-scale data integration to address complex, interconnected pollution sources.

Meanwhile, Topic 3's focus on sustainability and policy highlights the role of big data in translating scientific insights into actionable governance strategies.

This LDA analysis reveals that big data is not merely a technical tool but a catalyst for integrating diverse research domains, from environmental chemistry to urban planning. By enabling the synthesis of heterogeneous datasets, big data supports a more holistic understanding of water pollution dynamics, facilitating the development of adaptive, evidence-based solutions that balance ecological health with human needs. As global water challenges intensify due to climate change and urbanization, the role of big data in pollution reduction will only grow, underscoring the need for continued investment in data infrastructure and interdisciplinary collaboration.

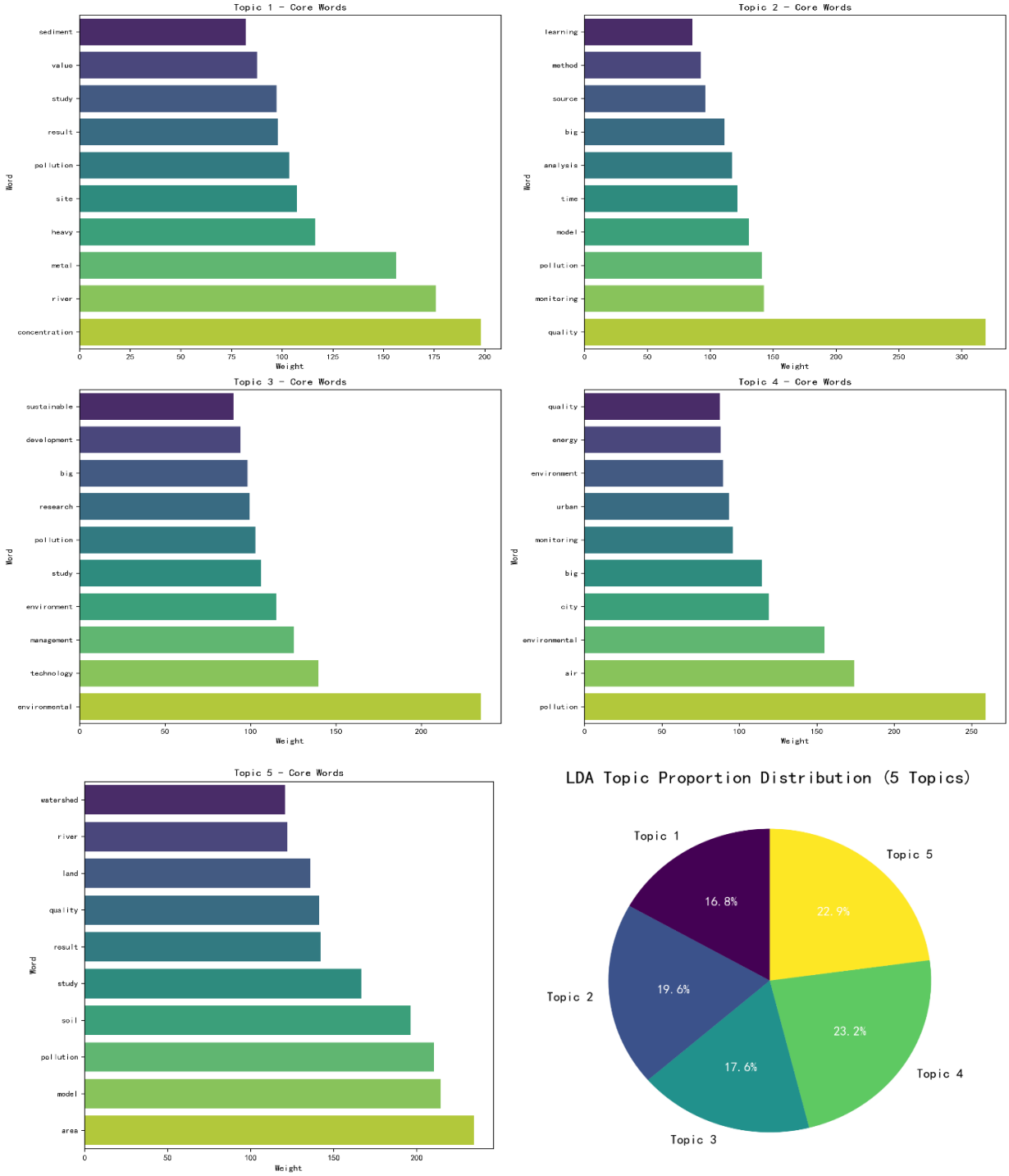

Figure 9. LDA topic

Figure 10. LDA topic proportion distribution

5 Conclusion

Despite significant advancements, big data applications in water pollution reduction face several challenges. Data heterogeneity and interoperability issues hinder the integration of diverse datasets (e.g., sensor data, remote sensing imagery, socioeconomic data) (McCabe et al., 2017). Additionally, data quality and accessibility remain problematic, particularly in developing regions where monitoring infrastructure is limited (Ahamad et al., 2024). Computational limitations, such as the need for advanced algorithms to process large volumes of unstructured data, also pose barriers (Wang et al., 2024).

Future research should focus on addressing these challenges through interdisciplinary collaboration, standardized data sharing platforms, and advances in artificial intelligence (AI) and machine learning. For example, federated learning, as proposed by Arshad et al. (2024), can enhance data privacy while enabling collaborative analysis of distributed pollution datasets. Integrating multi-omics data (e.g., genomics, metabolomics) with environmental big data may further improve pollutant detection and risk assessment (Charrasse et al., 2023). Additionally, expanding big data applications to urban agglomerations and transboundary water systems will be critical for addressing complex, large-scale pollution problems (Fang et al., 2016).

Big data has emerged as a powerful tool for reducing water pollution, enabling enhanced monitoring, precise source identification, accurate risk assessment, and optimized mitigation strategies. The integration of remote sensing, sensor networks, and computational models provides a comprehensive understanding of pollution dynamics, supporting evidence-based decision-making. While challenges related to data integration, quality, and computational capacity persist, ongoing advancements in AI, machine learning, and data sharing will further unlock the potential of big data in water pollution management. By leveraging big data technologies, stakeholders can work towards more sustainable water resource management and protect ecological and human health.

References

Ahamad, H., Alam, O., Sultana, N., & Misbahuzzaman, K. (2024). Current environmental management practices in major manufacturing industries of Chattogram, Bangladesh. Environment Development and Sustainability. https://doi.org/10.1007/s10668-024-05697-w

Amador, P., Gherardi, V., Fuentes-Edfuf, Y., Martinez-Megias, C., & Rico, A. (2025). Thinking big! Landscape-scale evaluation of pesticide pollution and ecological risks in a protected mediterranean wetland. Environmental Pollution, 384, 126918. https://doi.org/10.1016/j.envpol.2025.126918

Arshad, J., Irtisam, A., Arif, T., Rasheed, M. S., Chauhdary, S. T., Rahmani, M. K. I., & Almajalid, R. (2024). A federated learning model for intelligent cattle health monitoring system using body area sensors and IoT. Egyptian Informatics Journal, 27, 100488. https://doi.org/10.1016/j.eij.2024.100488

Bolca, M., Turkyilmaz, B., Kurucu, Y., Altinbas, U., Esetlili, M. T., & Gulgun, B. (2007). Determination of impact of urbanization on agricultural land and wetland land use in balcovas' delta by remote sensing and GIS technique. Environmental Monitoring and Assessment, 131(1–3), 409–419. https://doi.org/10.1007/s10661-006-9486-0

Bojanowski, D., Orlinska-Wozniak, P., Wilk, P., Jakusik, E., & Szalinska, E. (2024). Spatial and temporal changes in nutrient source contribution in a lowland catchment within the Baltic Sea region under climate change scenarios. Water Resources Research, 60(5), e2023WR034979. https://doi.org/10.1029/2023WR034979

Charrasse, S., Poquillon, T., Saint-Omer, C., Pastore, M., Bordignon, B., Frye, R. E., Reynes, C., Racine, V., & Aouacheria, A. (2023). Quantitative assessment of mitochondrial morphology relevant for studies on cellular health and environmental toxicity. Computational and Structural Biotechnology Journal, 21, 5609–5619. https://doi.org/10.1016/j.csbj.2023.11.015

Elewa, K., Tawfic, A. F., Tarek, M., Al-Sagheer, N. A., & Nagy, N. M. (2025). Utilizing metakaolin and siliceous waste from the alum industry to create geopolymer adsorbent for the removal of certain heavy metals. Scientific Reports, 15(1),

27484. https://doi.org/10.1038/s41598-025-10800-w

Fang, C., Liu, H., & Li, G. (2016). International progress and evaluation on interactive coupling effects between urbanization and the eco-environment. Journal of Geographical Sciences, 26(8), 1081–1116. https://doi.org/10.1007/s11442-016-1317-9

Farooq, M., Khan, U., Adnan, M., Younas, W., Qureshi, N. B., Yasir, Q. M., Cai, Q., & Chiu, M.-C. (2025). Global decline of aquatic and terrestrial insects driven by climate change and anthropogenic impacts: The interaction of multiple stressors and disruption of niche conservatism. Biological Conservation, 308, 111181. https://doi.org/10.1016/j.biocon.2025.111181

Kramm, G., Dlugi, R., Dollard, G. J., Foken, T., Molders, N., Müller, H., Seiler, W., & Sievering, H. (1995). On the dry deposition of ozone and reactive nitrogen species. Atmospheric Environment, 29(21), 3209–3231. https://doi.org/10.1016/1352-2310(95)00218-N

Lazartigues, A., Banas, D., Feidt, C., Brun-Bellut, J., Gardeur, J.-N., Le Roux, Y., & Thomas, M. (2013). Pesticide pressure and fish farming in barrage pond in northeastern France. Part III: how management can affect pesticide profiles in edible fish? Environmental Science and Pollution Research, 20(1), 126–135. https://doi.org/10.1007/s11356-012-0824-1

Li, F., Xiang, M., Liu, S., Lei, K., Shi, Z., Yang, J., & Li, Y. (2021). Life cycle assessment and fertilization scheme optimization of paddy field crops in South China. Journal of Cleaner Production, 325, 129339. https://doi.org/10.1016/j.jclepro.2021.129339

McCabe, M. F., Rodell, M., Alsdorf, D. E., Miralles, D. G., Uijlenhoet, R., Wagner, W., Lucieer, A., Houborg, R., Verhoest, N. E. C., Franz, T. E., Shi, J., Gao, H., & Wood, E. F. (2017). The future of Earth observation in hydrology. Hydrology and Earth System Sciences, 21(7), 3879–3914. https://doi.org/10.5194/hess-21-3879-2017

Muscillo, M., La Rosa, G., Carducci, A., Cantiani, L., & Marianelli, C. (1999). Molecular and biological characterization of poliovirus 3 strains isolated in Adriatic seawater samples. Water Research, 33(14), 3204–3212. https://doi.org/10.1016/S0043-1354(99)00028-7

Naboni, E., & Garcia, D. (2017). Sensitive and reactive architectural devices for regenerative design. Applications in Manaus Heat Islands and Surrounding Rain Forests. In L. Malinovska, V. Osadcuks, & V. Katerina (Eds.), Sustainable Synergies from Buildings to the Urban Scale (pp. 658–665). IOP Publishing. https://doi.org/10.1016/j.proenv.2017.03.146

Prishchepenko, D. V., Ryabchuk, D. V., Zhamoida, V. A., Sergeev, A. Y., Leontev, F. A., Grigoriev, A. G., Neevin, I. A., Budanov, L. M., & Kovaleva, O. A. (2023). Main trends and results of 300-years anthropogenic impact on the geological environment and ecosystem of the Eastern Gulf of Finland. Continental Shelf Research, 265, 105058. https://doi.org/10.1016/j.csr.2023.105058

Radulescu, V. (2020). Temperature distribution in a turbulent flow in the heat pollution phenomena. In Proceedings of the ASME 2020 Heat Transfer Summer Conference (HT2020). American Society of Mechanical Engineers. https://doi.org/10.1115/HT2020-8969

Schletterer, M., Fuereder, L., Kuzovlev, V. V., & Beketov, M. A. (2010). Testing the coherence of several macroinvertebrate indices and environmental factors in a large lowland river system (Volga River, Russia). Ecological Indicators, 10(6), 1083–1092. https://doi.org/10.1016/j.ecolind.2010.03.004

Skorobogatova, O. N. (2018). Taxonomic composition of phytoplankton in the Vakh River (Western Siberia). In 5th International Field Symposium West Siberian Peatlands and Carbon Cycle: Past and Present and the International Conference Carbon Balance of Western Siberian Mires in the Context of Climate Change (pp. 012017). IOP Publishing. https://doi.org/10.1088/1755-1315/138/1/012017

Subbotin, I., Briukhanov, A., & Vasilev, E. (2017). Factor analysis of environmental impact of manure utilization. In L. Malinovska & V. Osadcuks (Eds.), 16th International Scientific Conference: Engineering for Rural Development (pp. 625–629). Latvia University of Agriculture. https://doi.org/10.22616/ERDev2017.16.N124

Takata, H., Aono, T., Tagami, K., & Uchida, S. (2010). Concentration ratios of stable elements for selected biota in

Japanese estuarine areas. Radiation and Environmental Biophysics, 49(4), 591–601. https://doi.org/10.1007/s00411-010-0317-x

Wang, H., Chen, J., Ma, F., Zhu, M., Su, L., Liu, W., & Zhang, Y. (2023). Environmental computational toxicology for screening persistent, bio-accumulative, and toxic chemicals: Progress and perspectives. Chinese Science Bulletin-Chinese, 69(6), 688–702. https://doi.org/10.1360/TB-2023-0351

Xie, H., Zhang, Y., Choi, Y., & Li, F. (2020). A scientometrics review on land ecosystem service research. Sustainability, 12(7), 2959. https://doi.org/10.3390/su12072959

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